

CS166 HW1

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Problem 1) Exercise 2.8 from the textbook

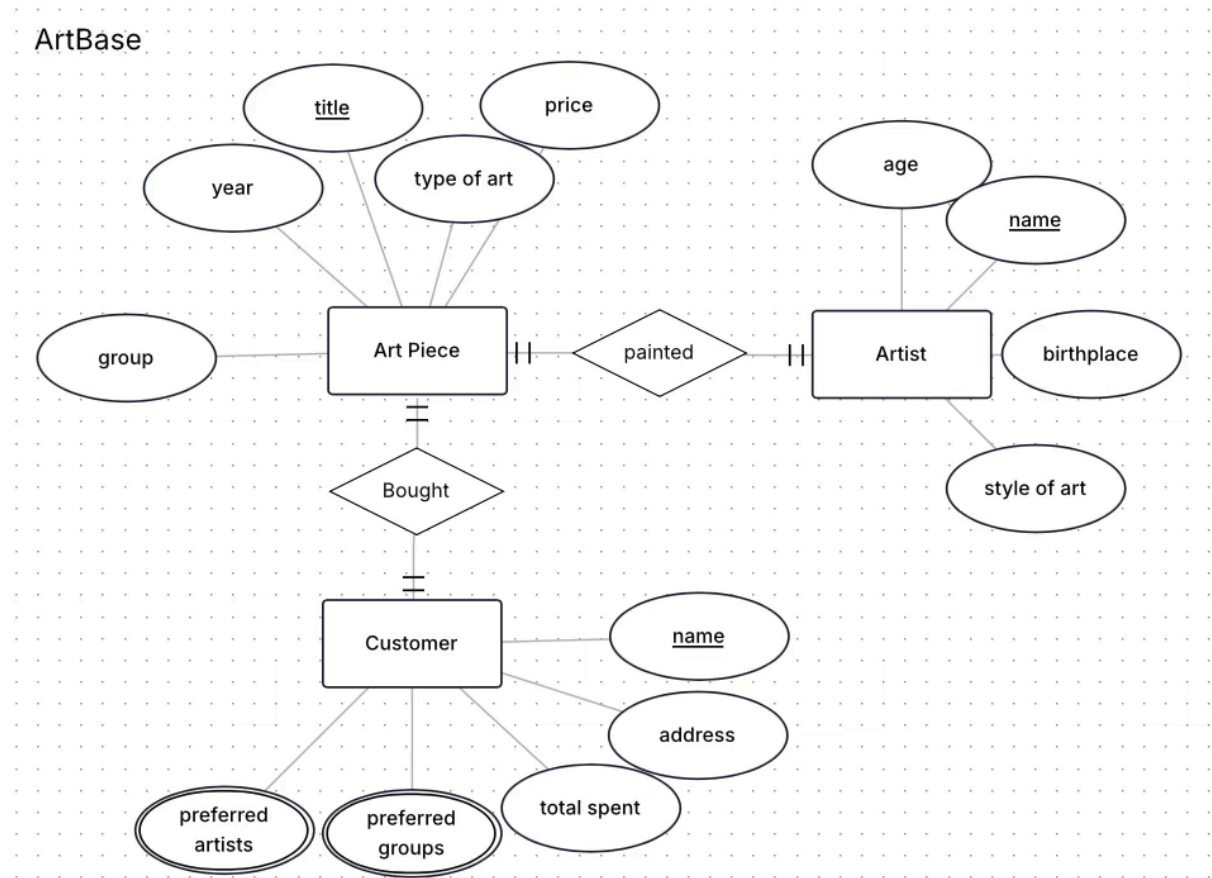


Figure 1: ArtBase database schema from the textbook, used in Exercise 2.8

Problem 2) Exercise 3.9; Q: 2-5

Questions are based off the following schema:

Emp(eid: integer, ename: string, age: integer, salary: real)

Works(eid: integer, did: integer, pcttime: integer)

Dept(did: integer, dname: string, budget: real, managerid: integer)

2. SQL to create proceeding relations, including appropriate version of all primary and foreign key integrity constraints.

```
CREATE TABLE Emp (  
  eid INTEGER PRIMARY KEY, -- Primary key is Employee ID  
  ename STRING, -- employee name  
  age INTEGER, -- employee age  
  salary REAL -- employee salary  
);
```

3. Define the Dept relationship in SQL so that every department is guaranteed a manager.

```
CREATE TABLE Dept (  
  did INTEGER PRIMARY KEY, // Primary key is Department ID  
  dname STRING, -- department name  
  budget REAL, -- department budget  
  managerid INTEGER, -- manager ID  
  FOREIGN KEY (managerid) REFERENCES Emp(eid) /* Assigning managerid as a foreign key  
referencing Emp relation */  
);
```

4. SQL to add John Doe as an employee with eid = 101, age = 32, and salary = 15,000 to the Employee relation, and to assign him to the Sales department.

```
INSERT INTO Emp VALUES (101, 'John Doe', 32, 15000);
```

5. SQL to give every employee a 10% raise.

```
UPDATE Emp SET salary = salary * 1.10;
```

SQL commands for Problem 3

```
CREATE TABLE Airplane(  
    Company CHAR(20),  
    reg_no CHAR(20) PRIMARY KEY,  
);  
  
CREATE TABLE Model(  
    model_no CHAR(20) PRIMARY KEY,  
    Seat-capacity INTEGER,  
    weight INTEGER,  
    fuel INTEGER  
);  
  
-- Relationship between Airplane and Model, Type. Every airplane is a model  
CREATE TABLE Type(  
    reg_no CHAR(20) PRIMARY KEY,  
    model_no CHAR(20),  
    FOREIGN KEY (reg_no) REFERENCES Airplane(reg_no),  
    FOREIGN KEY (model_no) REFERENCES Model(model_no)  
);  
  
CREATE TABLE Technician(  
    ssn CHAR(15) PRIMARY KEY,  
    salary INTEGER,  
    Last-name CHAR(20),  
    address CHAR(30),  
    Phone_num CHAR(15),  
    FOREIGN KEY (ssn) REFERENCES Employee(ssn)  
);  
  
-- Relationship between Airplane and Technician, Can-Fix, 1-1 relationship  
CREATE TABLE Can_Fix(  
    reg_no CHAR(20) PRIMARY KEY NOT NULL,  
    ssn CHAR(15) NOT NULL,  
    FOREIGN KEY (reg_no) REFERENCES Airplane(reg_no),  
    FOREIGN KEY (ssn) REFERENCES Technician(ssn)  
);  
  
CREATE TABLE Employee(  
    ssn CHAR(11) PRIMARY KEY,  
    union_mem_no INTEGER  
);  
  
-- Have a foreign key of level  
CREATE TABLE Traffic-Controller(  
    age INTEGER,  
    Years-exp INTEGER  
);  
  
-- ISA relationship between Technician, employee and Traffic Controller  
CREATE TABLE Technician_ISA(  
    ssn CHAR(11) PRIMARY KEY,  
    FOREIGN KEY (ssn) REFERENCES Employee(ssn)  
);
```

```
CREATE Table Exam(  
    duration INTEGER,  
    date DATE,  
    level CHAR(20) NOT NULL  
);
```

-- Relationship between traffic controller and exam, must pass exam to be a traffic controller

```
CREATE TABLE Passed(  
    ssn CHAR(11) PRIMARY KEY NOT NULL,  
    level CHAR(20) NOT NULL,  
    FOREIGN KEY (ssn) REFERENCES Traffic-Controller(ssn),  
    FOREIGN KEY (level) REFERENCES Exam(level)  
);
```

```
CREATE TABLE Test(  
    name CHAR(20),  
    FAA_no CHAR(20) PRIMARY KEY,  
    Max-score INTEGER  
);
```

-- Test_info is a relationship between Technician, Airplane, and Test with attribute hours, date, score

```
CREATE TABLE Test_info(  
    ssn CHAR(11),  
    reg_no CHAR(20),  
    FAA_no CHAR(20),  
    hours INTEGER,  
    date DATE,  
    score INTEGER,  
    PRIMARY KEY (ssn, reg_no, FAA_no),  
    FOREIGN KEY (ssn) REFERENCES Technician(ssn),  
    FOREIGN KEY (reg_no) REFERENCES Airplane(reg_no),  
    FOREIGN KEY (FAA_no) REFERENCES Test(FAA_no)  
);
```